
```
clc;
clear;
close all;

% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;

% Message Signal
m = cos(2*pi*fm*t);

% Carrier Signal
c = cos(2*pi*fc*t);

% DSB-SC Modulation
dsb = m .* c;

% VSB Filter
[b,a] = butter(4,fc/(fs/2));
vsb = filter(b,a,dsb);

% Coherent Demodulation
demod = vsb .* (2*cos(2*pi*fc*t));

% Message Recovery
[b1,a1] = butter(4,1000/(fs/2));
rec = filter(b1,a1,demod);

% Plot
figure('Name','VSB Modulation','NumberTitle','off');

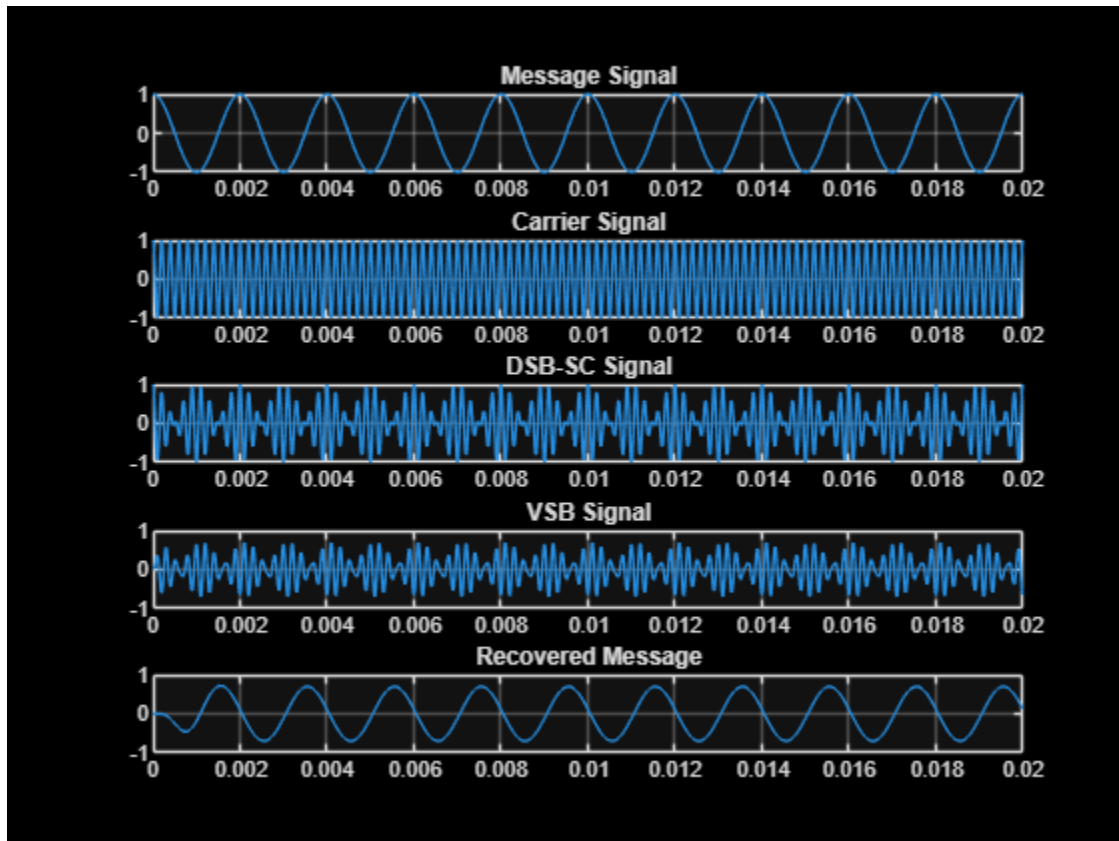
subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

subplot(5,1,3);
plot(t,dsb);
title('DSB-SC Signal');
grid on;

subplot(5,1,4);
plot(t,vsb);
title('VSB Signal');
grid on;
```

```
subplot(5,1,5);  
plot(t,rec);  
title('Recovered Message');  
grid on;
```



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